AH_G474 Header Board V0.1

Development header board based on the STM32G474RBT6 microcontroller (Arm Cortex-M4, 170 MHz) with USB Type-C, SWD and an on-board CR1220 battery holder

170 MHz	128 KB / 128 KB	USB Type-C	65 × 26 mm
Arm Cortex-M4 core, STM32G474RBT6	Flash / SRAM, on the STM32G474RBT6	USB 2.0 Full-Speed, 5 V supply input	2 × 25-pin 2.54 mm headers, 43 GPIO

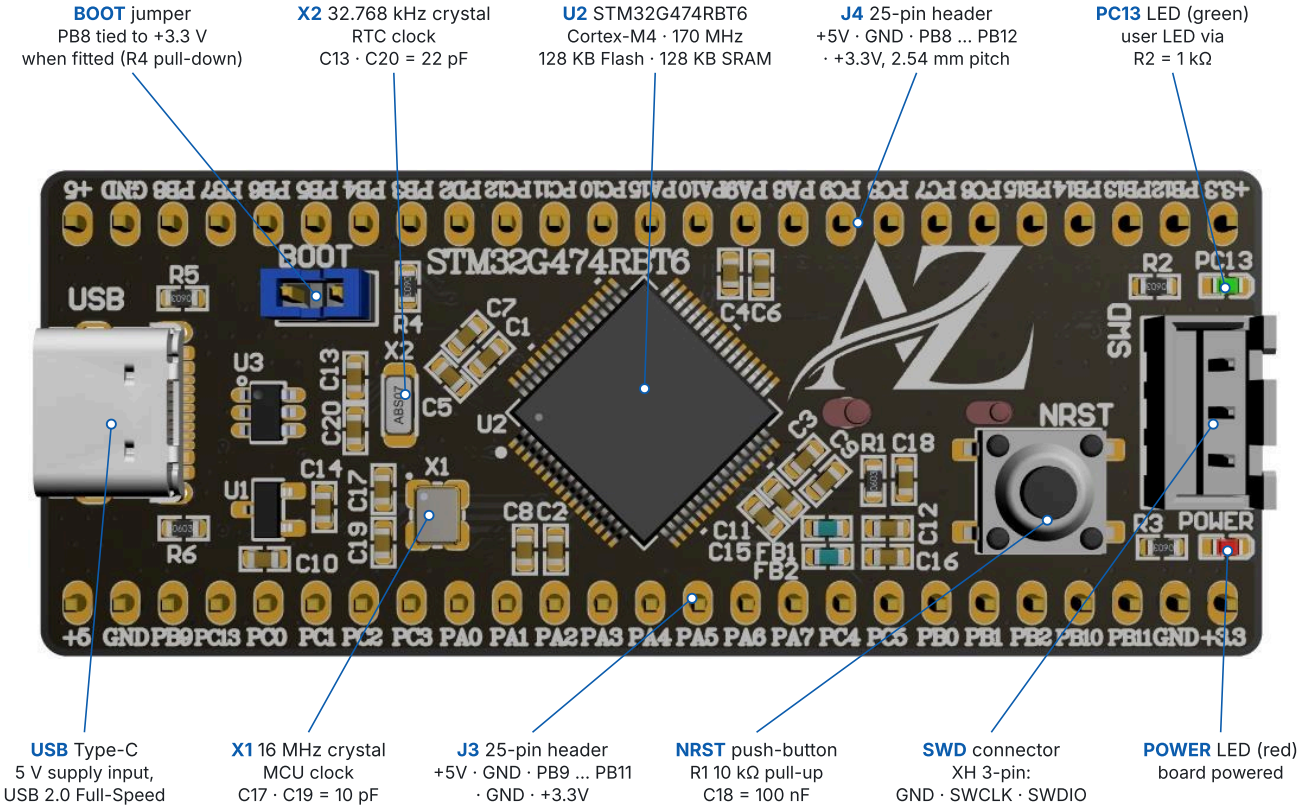


Figure 1. Top view of the AH_G474 header board (PCB render) with the user-relevant items called out. Component reference designators match the silkscreen and the schematic.

Header Pinout

The AH_G474 brings 43 GPIO lines of the STM32G474RBT6 out to two 25-pin, 2.54 mm headers: J4 along the top edge and J3 along the bottom edge (USB Type-C connector at the left). On both headers pin 1 is the +5 V rail, pin 2 is GND and pin 25 is the +3.3 V rail; J3 has a second GND at pin 24. Pins needed by on-board functions (USB, SWD, crystals and reset) are not routed to the headers.



Figure 2. Header pinout seen from the top of the board, pin 1 at the left (USB Type-C end). Colours group the pins by port; the dot marks the two pins shared with an on-board part.

Pins at a glance

Group	J4 (top edge), pin numbers in brackets	J3 (bottom edge), pin numbers in brackets	Pins
Power	+5V (1) · GND (2) · +3.3V (25)	+5V (1) · GND (2, 24) · +3.3V (25)	7
Port A	PA15, PA10, PA9, PA8 (13–16)	PA0–PA7 (9–16)	12
Port B	PB8–PB3 (3–8) · PB15–PB12 (21–24)	PB9 (3) · PB0–PB2 (19–21) · PB10–PB11 (22–23)	16
Port C	PC12–PC10 (10–12) · PC9–PC6 (17–20)	PC13 (4) · PC0–PC3 (5–8) · PC4–PC5 (17–18)	14
Port D	PD2 (9)	—	1
Total	25 pins	25 pins	50

> **Shared pins.** PB8 is also connected to the BOOT jumper and to R4 (10 kΩ pull-down); PC13 also drives the green user LED through R2 (1 kΩ). Keep this loading in mind when using them as I/O (see 6.1 and 7.1). Of the 50 header pins, 43 are GPIO and 7 are power (2 × +5V, 2 × +3.3V, 3 × GND).

1. Description

The AH_G474 is a header-style development board built around the STM32G474RBT6 microcontroller. It is designed for compact dimensions, broad access to the microcontroller's capabilities and use in professional and control projects. The board is supplied with 5 V from the USB Type-C connector, and an AP2120N-3.3 regulator provides the 3.3 V rail for the MCU and the ADC reference. A 16 MHz crystal clocks the MCU, a 32.768 kHz crystal clocks the RTC and a CR1220 cell on the bottom side backs up VBAT. The I/O is brought out to two 25-pin headers, and the board is programmed and debugged over SWD.

Features

- STM32G474RBT6: Arm Cortex-M4 up to 170 MHz, 128 KB Flash, 128 KB SRAM
- 16 MHz crystal for the MCU and 32.768 kHz crystal for the RTC
- USB Type-C connector, USB 2.0 Full-Speed
- SWD programming and debug interface (3-pin XH connector)
- Two 25-pin 2.54 mm headers (J4, J3) with 43 GPIO, +5V, +3.3V and GND
- UART / SPI / I²C / FDCAN, ADC / DAC, advanced timers and PWM
- On-board CR1220 battery holder for the RTC backup supply (VBAT)
- NRST push-button, BOOT jumper, PC13 user LED and POWER LED
- Compact 65 × 26 mm board

Applications

- Professional and industrial control projects
- PWM and timer-based control using the advanced timers
- Analog measurement and signal generation with the ADC and DAC
- Communication nodes using UART, SPI, I²C and FDCAN
- Firmware development and evaluation of the STM32G474RBT6

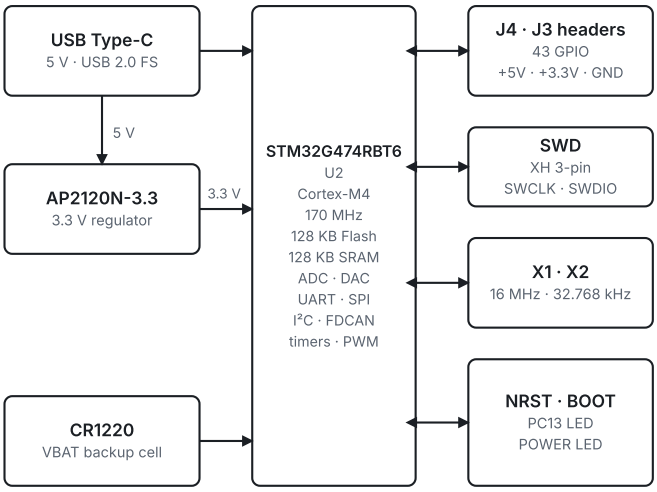


Figure 3. Functional block diagram.

2. Absolute Maximum Ratings

Stresses beyond these values may cause permanent damage. Operation at these limits is not implied.

Parameter	Min	Max	Unit	Note
Current drawn from the board by external loads	—	50	mA	Drawing more than 50 mA from the board damages the 3.3 V regulator (AP2120N-3.3).

Absolute maximum ratings of the STM32G474RBT6 pins and supplies, and of the AP2120N-3.3, are those given in the manufacturers' datasheets.

3. Recommended Operating Conditions

Parameter	Min	Typ	Max	Unit	Note
USB input voltage (USB Type-C)	—	5	—	V	Board supply input
Voltage on the +5V pins	—	5	—	V	J4 pin 1 and J3 pin 1
3.3 V rail (MCU, +3.3V pins)	—	3.3	—	V	From the AP2120N-3.3 regulator
ADC reference voltage VREF+	—	3.3	—	V	
Load current drawn from the board	—	—	50	mA	Higher current damages the regulator
CPU frequency	—	—	170	MHz	Arm Cortex-M4 of the STM32G474RBT6
RTC backup battery	CR1220 coin cell				On VBAT, holder on the bottom side

Values shown as "—" are not specified in the board documentation.

4. Electrical Characteristics

Board-level figures from the AH_G474 specification (Rev V0.1); tolerances are not specified.

Symbol	Parameter	Condition	Min	Typ	Max	Unit
V_{USB}	USB input voltage	USB Type-C connector		5		V
V_{5V}	+5V header voltage	+5V pins		5		V
V_{3V3}	+3.3V header voltage	+3.3V pins		3.3		V
V_{DD}	MCU supply voltage			3.3		V
V_{REF+}	ADC reference voltage			3.3		V
f_{X1}	Main crystal frequency	X1, MCU clock		16		MHz
f_{X2}	RTC crystal frequency	X2, RTC clock		32.768		kHz
f_{CPU}	CPU frequency	Cortex-M4			170	MHz
I_{LOAD}	Load current drawn from the board	all external loads			50	mA

4.1 Device summary

Parameter	Value	Description
MCU	STM32G474RBT6	Arm Cortex-M4, up to 170 MHz
Memory	128 KB Flash · 128 KB SRAM	On the MCU
Regulator IC	AP2120N-3.3	3.3 V rail
RTC battery	CR1220	Connected to VBAT
USB interface	USB 2.0 Full-Speed	USB Type-C connector
Peripherals	UART / SPI / I²C / FDCAN · ADC / DAC · advanced timers · PWM	I/O on J4 and J3

5. Pin and Pad Description

Name	Location	Type	Description
+5V	J4 pin 1, J3 pin 1	Power	5 V rail (USB Type-C supply voltage).
+3.3V	J4 pin 25, J3 pin 25	Power out	3.3 V rail from the AP2120N-3.3 regulator (MCU supply, VREF+ = 3.3 V). Total external load: 50 mA max.
GND	J4 pin 2, J3 pins 2 and 24, SWD pin 1	Ground	Common return for all supplies and signals.
PA0–PA10, PA15	J3 pins 9–16 (PA0–PA7) J4 pins 16–14 (PA8–PA10), pin 13 (PA15)	I/O	12 GPIO of port A.
PB0–PB15	J3 pins 19–21 (PB0–PB2), 3 (PB9), 22–23 (PB10–PB11) J4 pins 8–3 (PB3–PB8), 24–21 (PB12–PB15)	I/O	16 GPIO of port B. PB8 is shared with the BOOT jumper and R4.
PC0–PC13	J3 pins 5–8 (PC0–PC3), 17–18 (PC4–PC5), 4 (PC13) J4 pins 20–17 (PC6–PC9), 12–10 (PC10–PC12)	I/O	14 GPIO of port C. PC13 also drives the green user LED (R2 = 1 kΩ).
PD2	J4 pin 9	I/O	1 GPIO of port D.
SWD	XH 3-pin connector, right edge; pads DIO / CLK / GND on the bottom side	Debug	Pin 1 GND, pin 2 SWCLK, pin 3 SWDIO.
USB	USB Type-C connector, left edge	USB	USB 2.0 Full-Speed interface; 5 V supply input.
NRST	Push-button, right of the MCU	Input	Pulls the NRST net to GND (reset). R1 = 10 kΩ pull-up to +3.3 V, C18 = 100 nF.
BOOT	2-pin jumper, top-left next to the USB connector	Config	Fitted: PB8 tied to +3.3 V. Open: PB8 pulled to GND by R4 (10 kΩ).
PC13	Green LED, top-right edge	Indicator	User LED, on while PC13 is driven high.
POWER	Red LED, bottom-right edge	Indicator	On while the board is powered.
VBAT	CR1220 holder, bottom side	Power in	RTC backup supply from the coin cell.

6. Functional Description

6.1 Status indicators

Condition	PC13 (green)	POWER (red)
Board powered, PC13 driven high	On	On
Board powered, PC13 driven low or not driven	Off	On
No supply	Off	Off

The user LED (green, 0603) is connected from PC13 through R2 (1 kΩ) to GND; PC13 is also available on J3 pin 4.

6.2 Clock, reset and boot circuits

Circuit	Part	Connection	Components
Main clock	X1, 16 MHz crystal	OSC_IN / OSC_OUT; X1 pins 2 and 4 to GND	C17, C19 = 10 pF to GND
RTC clock	X2, 32.768 kHz crystal	OSC32_IN / OSC32_OUT	C13, C20 = 22 pF to GND
Reset	NRST push-button	NRST net, button to GND	R1 = 10 kΩ to +3.3 V, C18 = 100 nF to GND
Boot	BOOT jumper	+3.3 V to PB8	R4 = 10 kΩ from PB8 to GND
User LED	Green LED	PC13 to LED to GND	R2 = 1 kΩ in series

6.3 Power and backup supply

- **USB supply:** the board is supplied with 5 V from the USB Type-C connector; the 5 V rail is also brought out to the +5V pins.
- **3.3 V rail:** the AP2120N-3.3 regulator supplies the MCU, VREF+ and the +3.3V pins.
- **Load limit:** drawing more than 50 mA from the board damages the regulator.
- **Backup battery:** the CR1220 holder on the bottom side is connected to VBAT (RTC backup).

7. Configuration

7.1 Boot selection (BOOT jumper, PB8)

BOOT jumper	PB8 level	Connection
Open (not fitted)	Low	PB8 pulled to GND by R4 (10 kΩ)
Fitted	High (3.3 V)	PB8 tied to +3.3 V

PB8 is the BOOT0 pin of the STM32G474RBT6; the boot source selected by its level is defined by the device's boot configuration (see the ST reference manual RM0440).

> **Do not drive PB8 low while the BOOT jumper is fitted.** With the jumper fitted PB8 is connected directly to +3.3 V (no series resistor), so an output driven low on PB8 would load the 3.3 V rail.

7.2 SWD connection (XH 3-pin)

Pin	Signal	Pad label on the bottom side
1	GND	GND
2	SWCLK	CLK
3	SWDIO	DIO

7.3 RTC backup battery

Fit a CR1220 coin cell in the holder on the bottom side. It supplies the VBAT pin of the MCU (RTC backup).

8. Application Notes

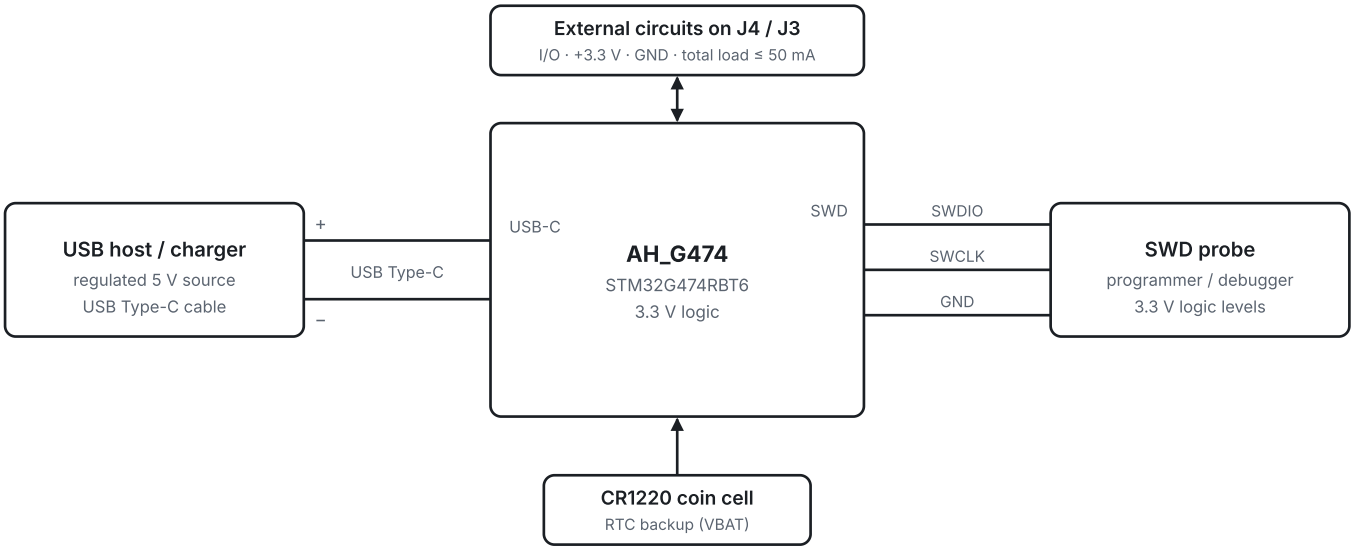


Figure 4. Typical connection. Power the board from USB Type-C, program it over SWD and keep the total external load below 50 mA.

8.1 Power supply

Power the board from a 5 V USB Type-C source. The AP2120N-3.3 regulator supplies all 3.3 V circuits and the +3.3V pins, so the total current available to external circuits is limited to 50 mA. For heavier loads use a separate supply and join the grounds.

8.2 Programming and debugging

Connect an SWD probe to the 3-pin XH connector (GND, SWCLK, SWDIO) or to the DIO / CLK / GND pads on the bottom side. The USB Type-C port provides a USB 2.0 Full-Speed interface to the MCU. The BOOT jumper selects the boot level of PB8 (see 7.1).

8.3 Using the headers

- Pin 1 of J4 and J3 is the +5V rail and pin 25 is the +3.3V rail; check the orientation before connecting a carrier board or wiring.
- PB8 and PC13 are shared with the BOOT jumper and the user LED (see Section 5).
- Pins used by USB, SWD, the crystals and reset are not available on the headers.

9. Mechanical Data

Board size	65.02 × 26.04 mm (nominal 65 × 26 mm), rectangular
Headers	J4 and J3, 1 × 25 pins each, 2.54 mm pitch
Row spacing	20.32 mm (0.8 in) between the J4 and J3 pin rows, centre to centre
USB connector	USB Type-C, left edge, top side
SWD connector	XH 3-pin, right edge, top side; DIO / CLK / GND pads labelled on the bottom side
Battery holder	CR1220, bottom side

Item	Side	Position
BOOT jumper	Top	top-left, next to USB
NRST button	Top	right of the MCU
PC13 LED (green)	Top	top-right edge
POWER LED (red)	Top	bottom-right edge
CR1220 holder	Bottom	near the centre (Fig. 6)

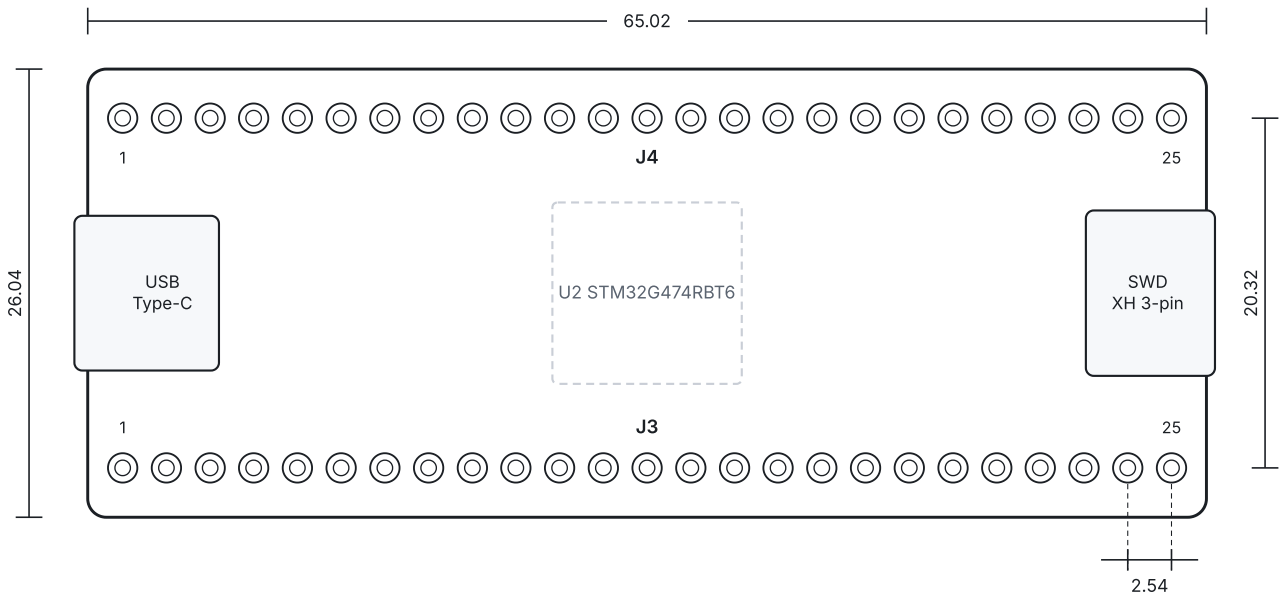


Figure 5. Outline and header positions, dimensions in mm. Connector and MCU outlines are schematic.

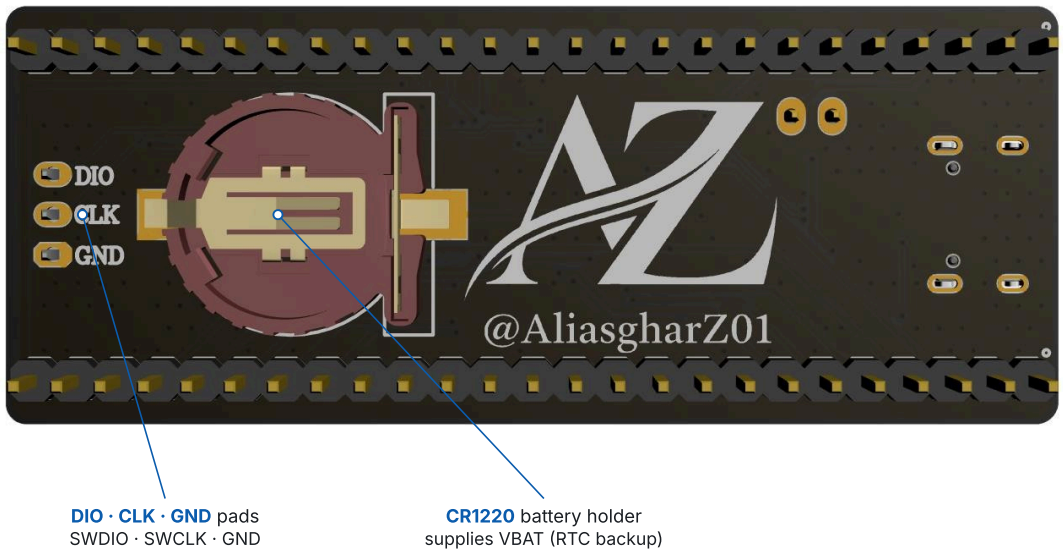


Figure 6. Bottom side of the board, seen from below: CR1220 battery holder and the DIO / CLK / GND pads (left–right mirrored compared with Figure 1).

10. Revision History

Doc rev	Date	Hardware	Changes
V0.1	November 2026	V0.1	Initial release.

@AliasgharZ01 — Information in this document is believed to be accurate at the time of publication and may change without notice. Parameters of the STM32G474RBT6 and the AP2120N-3.3 are those of the manufacturers' datasheets. Drawing more than 50 mA from the board damages the voltage regulator; the user is responsible for providing suitable external supplies and protection.